

Class 10 · Practice Test (Maths and Science)

Time: 1 hour · Maximum marks: 23 · Answer all questions. Show your working.

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- Q1. If the absolute refractive indices of two media X and Y are $\frac{6}{5}$ and $\frac{4}{3}$ respectively, then the refractive index of Y with respect to X will be: [1]
(A) $\frac{10}{9}$ (B) $\frac{9}{10}$ (C) $\frac{9}{8}$ (D) $\frac{8}{9}$
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- Q2. Assertion (A): If three vertices of a parallelogram taken in order are $(-1, -6)$, $(2, -5)$ and $(7, 2)$, then its fourth vertex is $(4, 1)$. [1]
Reason (R): Diagonals of a parallelogram bisect each other.
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.
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- Q3. (a) In common practice silver is recovered from silver nitrate solution by the use of copper metal. Name the type of reaction that takes place in this process and give the chemical equation of the reaction involved. [1]
(b) Name the method used for refining silver. [1]
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- Q4. A hall has a length of 9.75 m, breadth of 6.75 m and height of 5.25 m. What is the length of the longest unmarked ruler that can exactly measure the dimensions of the hall? [2]
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- Q5. A 15 V battery is connected in series with three resistors of $2\ \Omega$, $3\ \Omega$, and $5\ \Omega$. Calculate: [3]
(A) The total resistance in the circuit. [1]
(B) The total current supplied by the battery. [1]
(C) The voltage drop across the $3\ \Omega$ resistor. [1]
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- Q6. If α and β are the zeroes of the quadratic polynomial $2x^2 - 8x + 5$, then find the value of $(\alpha + \frac{1}{\beta}) \times (\beta + \frac{1}{\alpha})$. [3]
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- Q7. (a) Name the parts of hind-brain of the human brain. [$1\frac{1}{2}$] [2]
(b) Which part of the hind-brain controls involuntary actions such as blood pressure and salivation? [$\frac{1}{2}$]
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- Q8. In a rectangular park measuring $50\text{ m} \times 40\text{ m}$, the gram panchayat plans to construct a rectangular swimming pool in the middle, surrounded by a uniform-width grass strip throughout the park. Find the dimensions of the swimming pool, if the area of the grassy strip is 1184 m^2 . [5]
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- Q9. Case study: A person allowed a narrow beam of white light from the sun to enter a dark room through a small aperture and placed a glass prism in its path in such a manner that the beam falls on the face AB of the prism. A screen S is placed on the other side of the prism, facing AC. On turning the prism slowly, a beautiful band of colours is obtained on the screen. It is the spectrum of sunlight. [4]
- (a) Name the phenomenon due to which a prism splits the incident white light into a band of colours. [1]
- (b) State the reason of getting a band of seven colours in the above case. [1]
- (c) Draw a labelled ray diagram to show the formation of a rainbow. [2]
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